

# Olist E-Commerce Marketplace Analysis

What drives revenue — and what threatens scaling. A deep-dive into 100K orders across Brazil, 2016–2018.

BRAZIL · 100K ORDERS · 2016–2018

SQL · PYTHON · POWER BI

# The Marketplace Is Growing. But Is This Growth Healthy?

13.2M

BRL Revenue

Total across all delivered orders

96,478

Delivered Orders

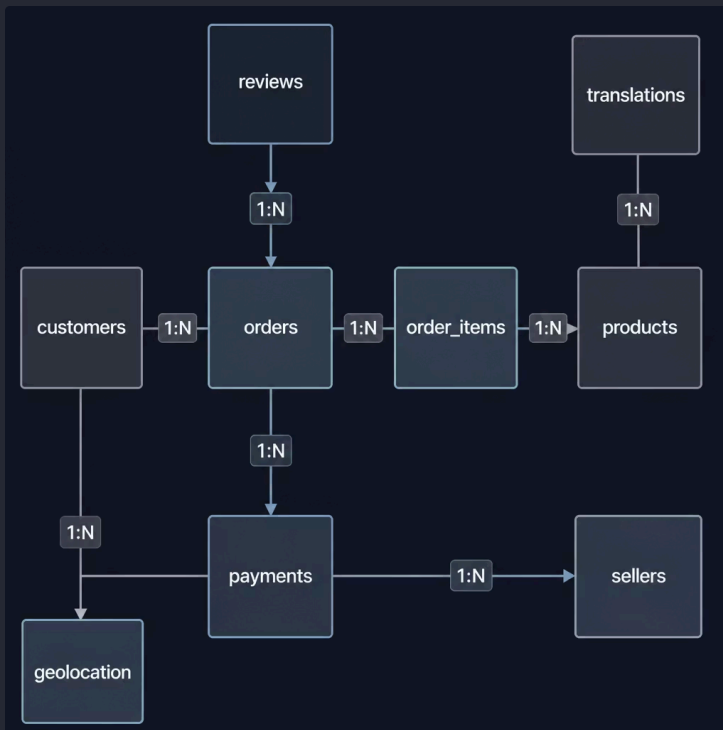
97% of total order volume

2,970

Active Sellers

Across all product categories

Three critical questions frame this analysis: Where does revenue really come from — and how concentrated is it? Do customers return, or must every sale be acquired from scratch? What breaks the customer experience, and who owns the problem?



# From 9 Raw CSV Files to an Executive Dashboard

## The Data Pipeline

Nine related CSV tables (~100K orders) were modeled into a relational schema (ERD), then processed through a structured pipeline:

📌 Scope was deliberate: all KPIs computed on **delivered orders only** — 97% of volume, excluding cancelled and undelivered orders.

- ➔ **Clean** — pandas / Power Query preprocessing
- ➔ **Model** — SQLite with window functions and CTEs
- ➔ **Validate** — 12 cross-validated KPIs
- ➔ **Visualize** — 3-page Power BI dashboard with DAX measures

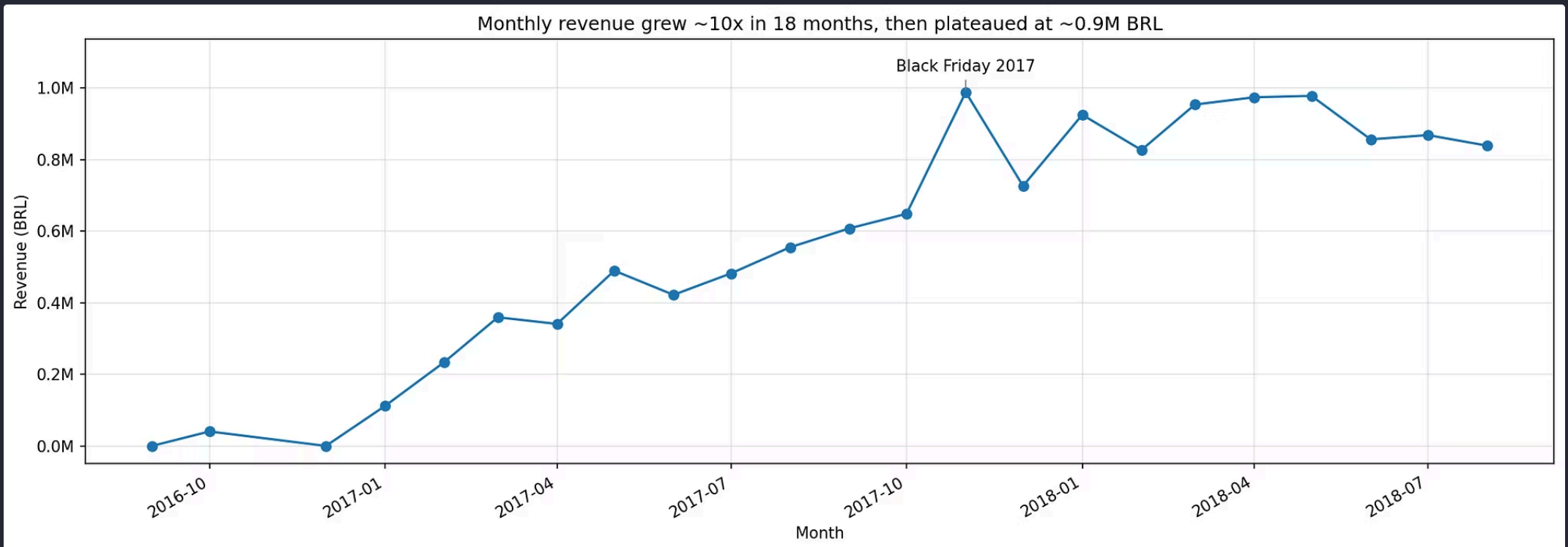
# Growth Is Bought, Not Built

## Retention Is Critically Low

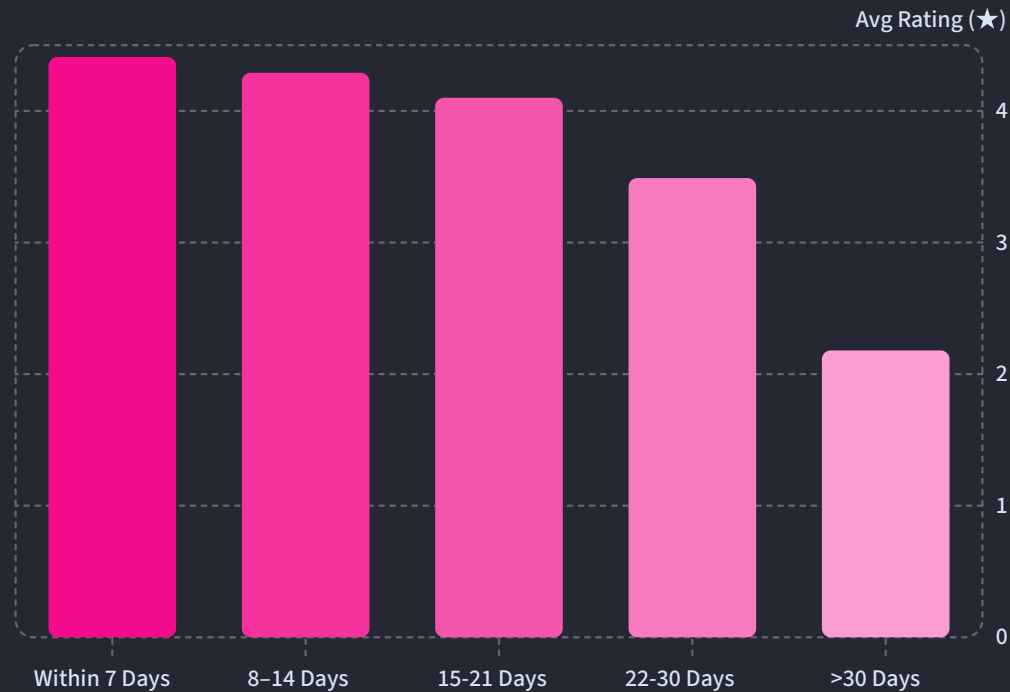
Only **3.0%** of customers ever place a second order — averaging just **1.03 orders per customer** across the entire dataset.

Black Friday 2017 delivered record sales, yet only **1.9%** of that cohort returned — the lowest repeat rate of any month in the dataset.

⚠ Without returning customers, every 1 BRL of revenue requires fresh acquisition spend. Growth lasts only as long as you pay for traffic.



# Customer Satisfaction Is Set by Logistics, Not the Product



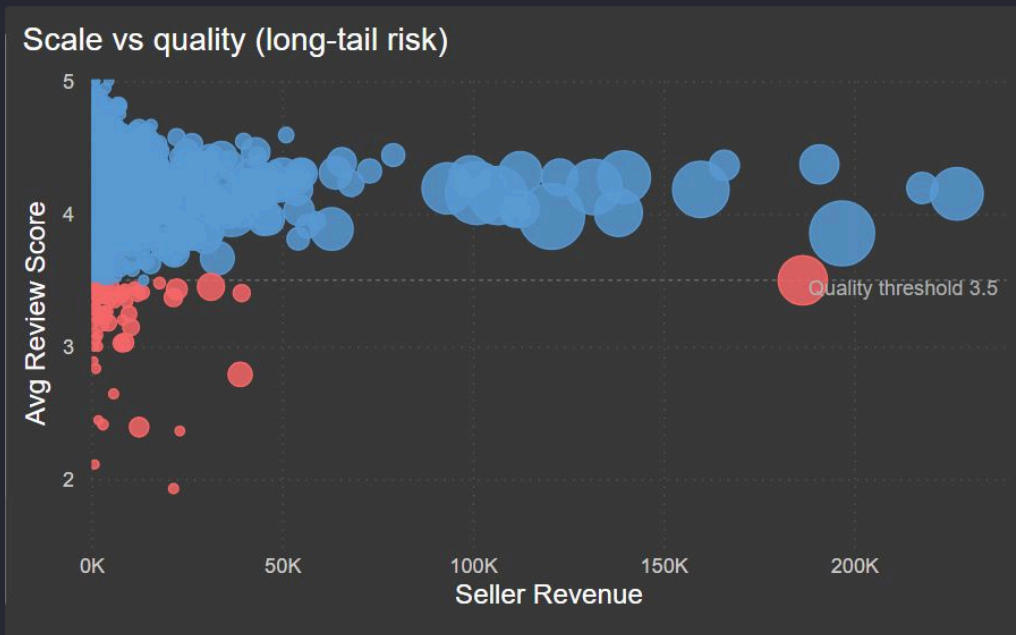
## Delivery Speed Drives Ratings

Customers who receive orders within 7 days rate 4.41★ on average. Ratings hold above 4.1★ up to 21 days — then satisfaction collapses: 3.49★ at 22–30 days, and just 2.18★ beyond 30 days.

The root cause is geographic: São Paulo averages **8.3 days** delivery, while northern states take **26–27 days** — nearly 3× slower.

Logistics, not product quality, is the primary lever for customer satisfaction improvement.

# Big Sellers Make the Revenue. Small Sellers Make the Risk.



## Concentration Confirmed

Across 2,970 active sellers, the classic 80/20 rule holds: the **top 20%** generate **82.3%** of all revenue.

Yet **343 sellers** (1 in 9) carry an average rating below 3.5★ — almost exclusively in the long tail. Scale and quality are tightly correlated: no seller above 100K BRL revenue falls below 4.0★.

Extreme tail case: **1.93★ average rating** with **64% late deliveries**.

# Three Levers to Unlock Sustainable Growth



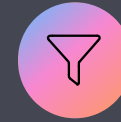
## Retain the Customer

Post-purchase loyalty program: a coupon for a complementary category, valid 30 days after delivery — conditional on a positive review.



## Deliver on Time

SLA with a hard **21-day limit** (the pain threshold found in data) and a **14-day operational target** for fulfillment teams.



## Filter Weak Sellers

Quality gate at onboarding + automatic alert when a seller's rating drops below 3.5★ — before they damage platform trust.

# What This Project Taught Me

## Cross-Validation Catches Real Bugs

Power BI flagged 348 "problem sellers"; SQL returned 343. The gap of 5 traced to a DAX bug: `BLANK < 3.5` evaluates TRUE, falsely flagging sellers with no reviews.

## Scope Is a Decision, Not an Assumption

Every KPI was deliberately computed on delivered orders only. The same metric with a different filter is a different number — both can be "correct."

## Next Steps

Cost data (CAC, margins) would convert directional recommendations into a business case with ROI — followed by pipeline automation.

[github.com/wp-pakulski/olist-ecommerce-analysis](https://github.com/wp-pakulski/olist-ecommerce-analysis)